

Project focus.
Students develop and evaluate machine-learning or deep-learning models for a specific clinical question, diagnostic target, or patient risk.
Model outputs are then passed to an LLM-based decision-support layer that summarizes the evidence for a clinician and prioritizes concerns from **urgent/high need** through **moderate**, **low**, or **no immediate need**.
The goal is a research prototype for clinician-facing synthesis— **not autonomous diagnosis or treatment**.

Research Datasets

Dataset	Data Types	Possible Clinical Targets / Risks	Access
All of Us – Registered Tier	Longitudinal EHR; inpatient/ED visits; diagnoses; medications; procedures; labs/measurements; surveys; physical measures; wearables	7/30-day readmission; ED return after discharge; repeat hospitalization; post-discharge mortality; utilization/length-of-stay questions; other longitudinal risk models	Registered Researcher Workbench access; individual onboarding/authorization required
PTB-XL	12-lead ECG waveforms; demographic/clinical metadata; cardiologist-derived ECG labels	Myocardial infarction; ST/T abnormalities; conduction disturbances; hypertrophy; normal vs. abnormal ECG; multi-label cardiac classification	Open via PhysioNet; CC BY 4.0
TCIA	CT, MRI, PET, pathology; collection-dependent segmentations, clinical outcomes, treatment data, radiomics/genomics	Cancer survival/mortality; recurrence or progression; treatment response; tumor phenotype; segmentation; radiogenomic prediction	Many collections public; access and license are collection-specific
ISIC Archive	Dermoscopic and clinical skin images; lesion diagnoses; some segmentation masks and patient/site metadata	Melanoma/malignancy detection; benign vs. malignant; multiclass lesion diagnosis; segmentation; subgroup/fairness analysis	Public research datasets; image/dataset license must be checked

Proposed Clinical-AI Pipeline

Clinical Dataset / Cohort → Task-Specific Predictive Model → Predictions + Risk Scores + Key Evidence → LLM Synthesis → Clinician-Facing Priority Summary

Urgent / High Moderate Low No Immediate Need

Model layer
Each student/team defines a clinically meaningful endpoint, builds leakage-safe train/validation/test cohorts, evaluates appropriate baselines and advanced models, and reports discrimination, calibration, error types, and subgroup performance where feasible.

LLM layer
The LLM should summarize *model outputs and supporting permitted context*, explain why an item is prioritized, surface uncertainty/conflicting signals, and preserve clinician review.
Restricted participant-level data should remain within the approved environment and must not be sent to external APIs unless the data-use terms explicitly allow it.

Research principle:
Start with the **clinical question**, not the model.
A publishable project should test whether a modeling or data-design choice improves clinically meaningful prediction, generalization, calibration, interpretability, or prioritization—not simply determine which algorithm has the highest accuracy.

Important Links

All of Us Registration	https://www.researchallofus.org/register/
PTB-XL (PhysioNet)	https://physionet.org/content/ptb-xl/1.0.3/
Cancer Imaging Archive	https://www.cancerimagingarchive.net/
ISIC Archive	https://www.isic-archive.com/
MedMNIST	https://medmnist.com/

